Allocation Patterns

- Map-Reduce
- Multi-Tier
- **Other Patterns**

Other Allocation Patterns

- There are several published deployment styles.

- Microsoft publishes a "Tiered Distribution" pattern, which prescribes a particular allocation of components in a multi-tier architecture to the hardware they will run on.

- IBM's WebSphere handbooks describe a number of what they call "topologies" along with the quality attribute criteria for choosing among them.
  - There are 11 topologies (specialized deployment patterns) described for Web-Sphere version 6, including the "single machine topology (stand-alone server)," "reverse proxy topology," "vertical scaling topology," "horizontal scaling topology," and "horizontal scaling with IP sprayer topology."

Other Allocation Patterns

- There are also published work assignment patterns. These take the form of often-used team structures. For example, patterns for globally distributed Agile projects include these:

  - **Platform**. In software product line development, one site is tasked with developing reusable core assets of the product line, and other sites develop applications that use the core assets.

  - **Competence center**. Work is allocated to sites depending on the technical or domain expertise located at a site. For example, user interface design is done at a site where usability engineering experts are located.

  - **Open source**. Many independent contributors develop the software product in accordance with a technical integration strategy. Centralized control is minimal, except when an independent contributor integrates his code into the product line.